



1. Introduction & Research Problem

- **High-grade serous ovarian carcinoma (HGSOC)** is the deadliest ovarian cancer.
- HGSOC affects ~600,000 women worldwide. Chemoresistance makes high-fidelity models critical for precision oncology^{1,2}.
- Patients frequently develop tumor spheroids within malignant effusions.
- **Patient-derived 3D organoids (PDOs)** from effusions are powerful models for precision drug screening to guide individualized therapies, and are ideal because they preserve native tumor architecture and patient heterogeneity^{1,3}.
- **Research Problem** = Transitioning cells from *in vivo* effusions to *in vitro* culture triggers transcriptomic shifts that could distort drug response data.

2. Research Question, Goal & Hypothesis

- **Research Question** = How does short-term *in vitro* culture (Day 0 vs. Day 6) alter the transcriptomic profile of patient-derived high-grade serous ovarian carcinoma (HGSOC) organoids?
- **Research Goals** = To map transcriptomic shifts during short-term culture (Day 0 vs. Day 6) and verify that patient-specific variation, not culture adaptation, drives model gene expression.
- **Hypothesis** = Transferring HGSOC cells from hypoxic malignant effusions to nutrient-rich culture triggers a transcriptomic shift, attenuating hypoxia and anti-apoptotic survival signaling while upregulating cell proliferation.

3. Experimental Design & Methodology

1. **Sample Preparation.** Isolated HGSOC cells from patient-derived malignant effusion to build 3D organoids.
2. **RNA Extraction.** Harvested RNA at Day 0 (*in vivo*) and Day 6 (*in vitro*).
3. **Preprocessing.** Quality-checked read counts and filter low-count genes.
4. **Differential Expression.** Modelled timepoint shifts using DESeq2 (Wald test, B-H FDR).
5. **QC & Clustering.** Applied VST for PCA and sample-distance heatmaps.
6. **Visualization & Interpretation.** Generated volcano and heatmap plots, identified key DEGs & mapped biological pathways via Enrichr.

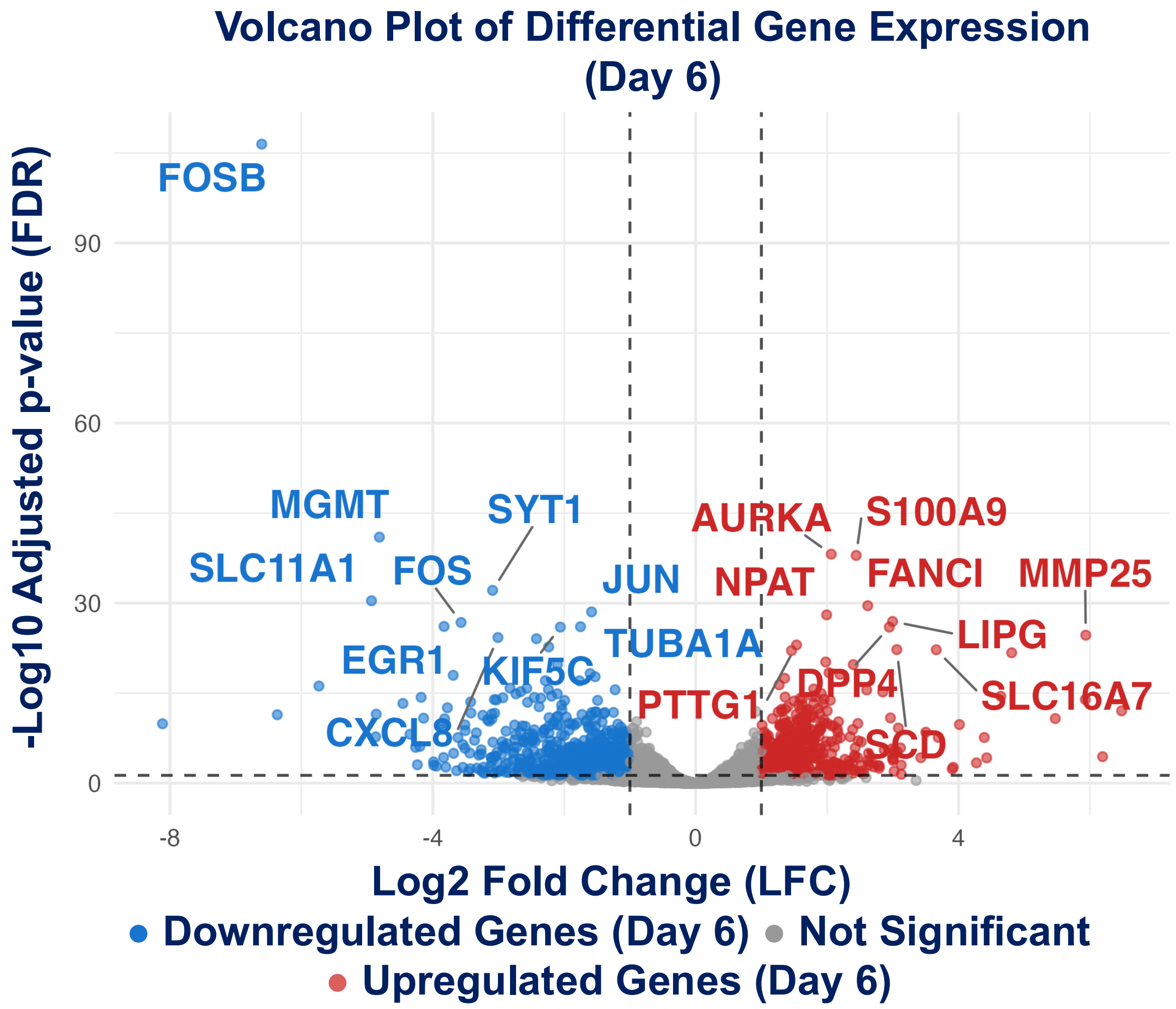


Figure 1. Volcano Plot of Differential Gene Expression (Day 6 vs. Day 0 baseline) across all HGSOC patient-derived organoids. Each point represents a gene, illustrating the transcriptomic shift (Days 0–6) from stress/hypoxia survival (downregulated genes (blue)) to active proliferation (upregulated genes (red)) after 6 days in culture.

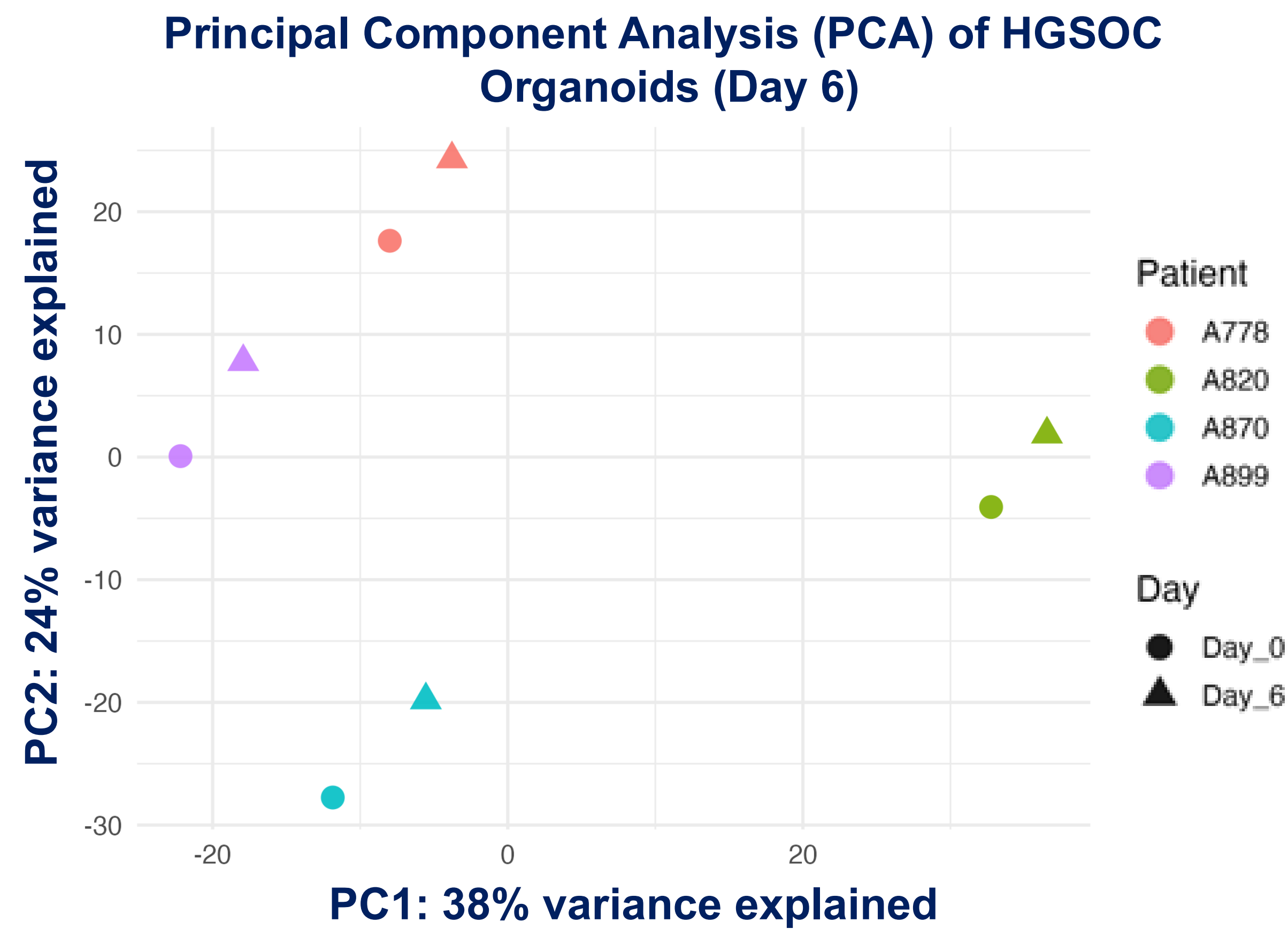


Figure 2. Principal Component Analysis (PCA) of HGSOC Organoids. PCA of variance-stabilized RNA-seq data showing global transcriptomic variation. **Strong clustering by patient indicates that patient-specific differences are the primary source of variation, confirming high fidelity of the organoid model.**

4. Results

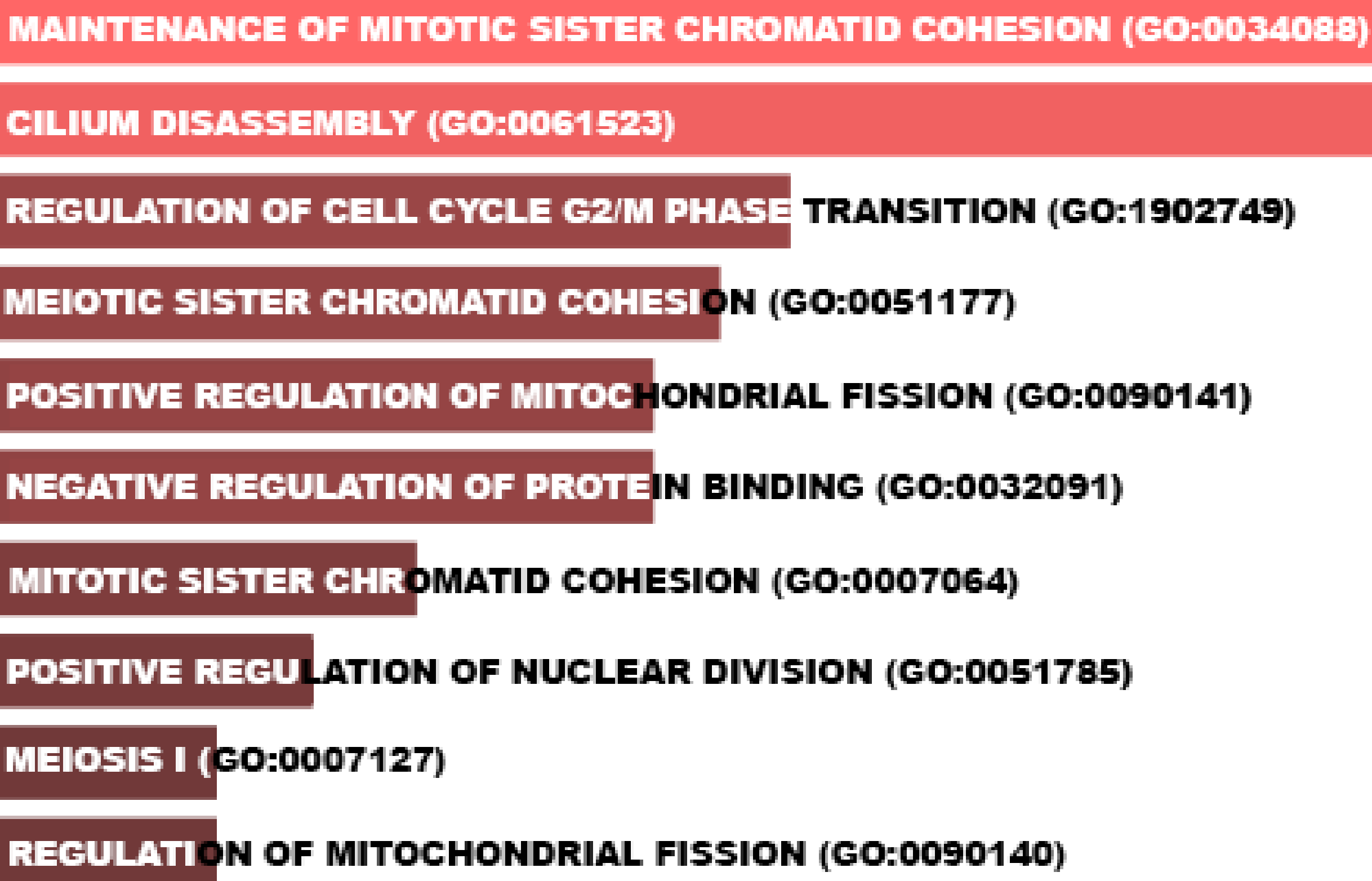


Figure 3. Gene Ontology Enrichment Analysis of Upregulated Genes at Day 6. **Top upregulated biological processes at Day 6**, showing exit from G0 (cilium disassembly), a resting state, and progression into active mitosis (chromosome cohesion, G2/M transition). Bars are ranked by adjusted *p*-value ($p < 0.05$). Plot generated using Enrichr (Chen et al., 2013; Xie et al., 2021).

Top 100 Differentially Expressed Genes (Day 6 v. Day 0)

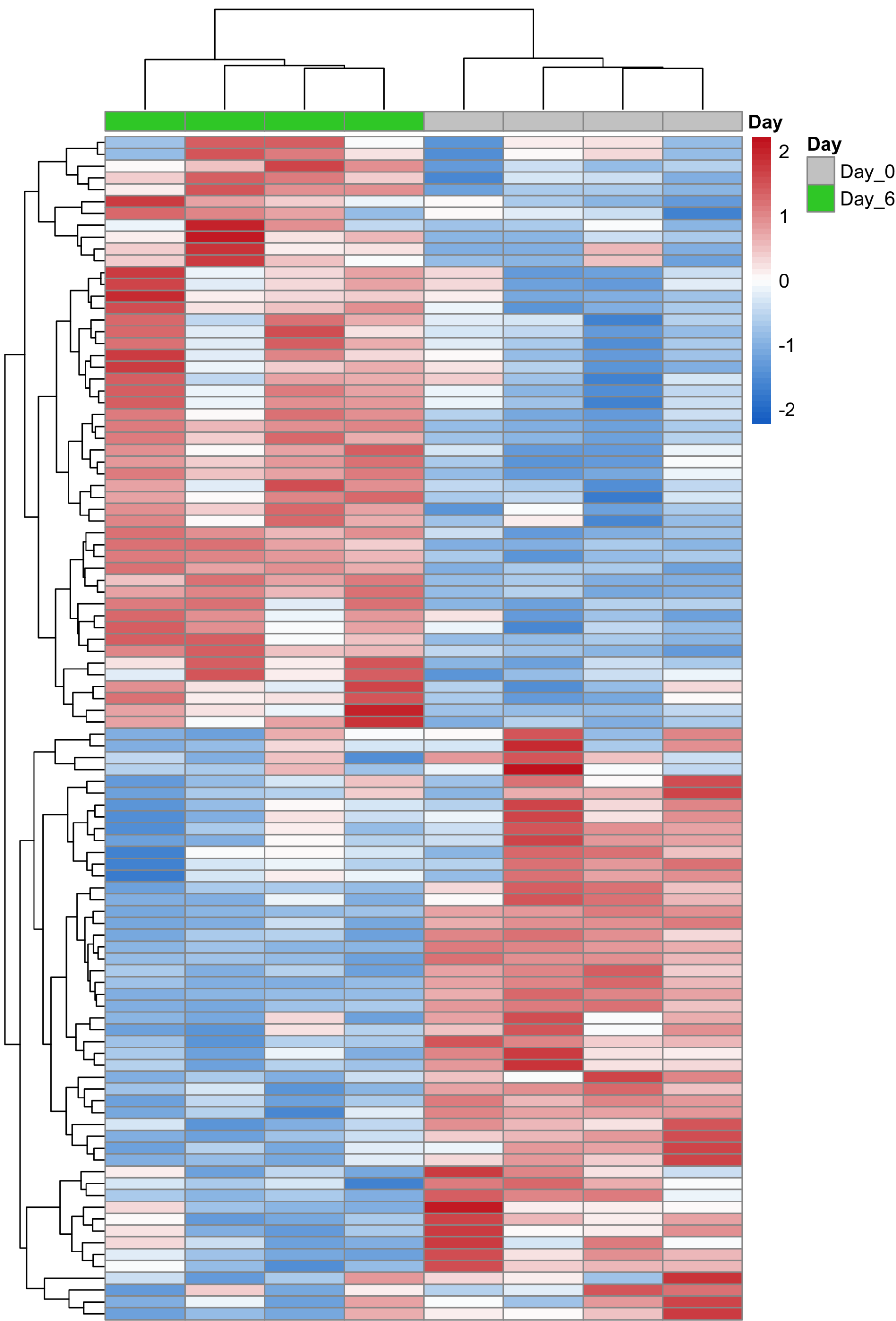


Figure 4. Heatmap of the Top 100 Differentially Expressed Genes (Day 6 vs. Day 0). Colour = Z Score. **Deep Red (+2) = Upregulated**, White (0) = Average Expression, Deep Blue (-2): Downregulated.

5. Biological Interpretation, Conclusion & Future Work

- Moving HGSOC organoids from *in vivo* to short-term *in vitro* culture induces a **systemic transcriptomic shift** from tissue stress to stable lab growth (Figure 4).
- **The top upregulated genes (smallest p_{adj}) (Figure 1) are known to drive cell cycle progression and DNA replication⁴⁻⁸ (Figure 3)**
- **The downregulated genes (Figure 1) are known to be associated with cell survival hypoxia, inflammation, and stress response pathways⁹⁻¹⁸.**
- Early-stage HGSOC organoids remain **reliable patient-specific models** despite transcriptomic shift (PC1 = 38%; Figure 2).
- **Hypothesis supported.** Transferring HGSOC cells from hypoxic malignant effusions to nutrient-rich culture triggers a transcriptomic shift from stress response to active proliferation.
- **Future studies** could involve reconstituting the tumor microenvironment, rescuing baseline stress genes, and creating an immunologically faithful platform for evaluating novel immunotherapies.

6. References

1. (Tadić et al., 2024); 2. (Ren et al. 2025); 3. (Qi et al., 2025); 4. (Blengini et al., 2025); 5. (Guarino Almeida et al., 2020), 6 (Vergani et al., 2025); 7. (Moriel-Carretero et al., 2017); 8 (Gao et al., 2003); 9. (NCBI, 2026); 10. (Ip and Davis, 1998); 11. (Wisdom et a., 1999); 12. (Zou and Zeng, 2023); 13.(Asokan and Bandapalli, 2021); 14. (Cambier et al., 2023); 15. (Rasras et al, 2024); 16. (Wang et al., 2021); 17. (Banerjee et al., 2025); 18. (Fang, 2024)(Chen et al. 2013)(Xie et al. 2021)

Acknowledgements

I would like to thank Professor Bolaños for her invaluable constructive feedback throughout this project.